

Definition

1. Top-Down Integration

Testing starts from the **top-level modules** (main control) and progresses downward to the lower-level modules.

- **Mechanism:** Uses **Stubs** (temporary dummy modules) to simulate the behavior of lower-level modules that are not yet developed.
- **Focus:** Validates high-level logic and architectural structure early.

2. Bottom-Up Integration

Testing starts with the **lowest-level modules** (atomic components) and moves upward to higher-level modules.

- **Mechanism:** Uses **Drivers** (temporary modules) to simulate the calling behavior of higher-level modules that are not yet developed.
- **Focus:** Validates critical low-level functions and utility components first.

3. Sandwich (Hybrid) Integration

A combination of Top-Down and Bottom-Up approaches. It tests the system from both ends simultaneously, meeting at a **middle layer**.

- **Mechanism:** Uses both Stubs (for the top) and Drivers (for the bottom).
- **Focus:** Balances the benefits of both methods, often focusing on a "target layer" where the most complex interactions occur.

4. Big Bang Integration

All modules are integrated and tested **simultaneously** after every single component has been developed.

- **Mechanism:** No stubs or drivers are usually required as the whole system is available.
- **Focus:** Suitable for small, simple systems. However, it is difficult to isolate defects because errors can originate from any module in the system.

Comparison

Feature	Top-Down	Bottom-Up	Sandwich	Big Bang
Direction	Top to Bottom	Bottom to Top	Both directions	All at once
Key Tool	Stubs	Drivers	Stubs & Drivers	None
Major Pro	Early prototype	Easy defect isolation	Balanced approach	Good for tiny projects
Major Con	Complex stubs	No early prototype	Higher overhead	Hard to find bug source
Focus	System Logic	Core Functions	Middle Layer	Final Assembly